

	$-9.65 \times 10^{-19} \text{ J}$
6(a)	Electric field strength in the parallel plate, $E = \frac{V}{d}$ Energy gained = work done by the electric force = electric force $\times$ distance moved $= qE \left(\frac{2d}{3} - \frac{d}{3} \right)$ $= q \left(\frac{V}{d} \right) \left(\frac{d}{3} \right)$ $= \frac{1}{3} qV$ (which is independent of the separation d)
6(b)	Gain in Kinetic Energy = Loss in Electric Potential Energy

$$\text{At } x = \frac{d}{3},$$

$$\frac{1}{2}mv^2 = \frac{1}{3}qV$$

$$v_1 = \sqrt{\frac{2qV}{3m}}$$

$$= \sqrt{\frac{2(2 \times 1.60 \times 10^{-19})(750)}{3 \times 4u}}$$

$$= 1.581 \times 10^5$$

$$\approx 1.58 \times 10^5 \text{ m s}^{-1}$$

$$\text{At } x = \frac{2d}{3},$$

$$\frac{1}{2}mv^2 = \frac{2}{3}qV$$

$$v_2 = \sqrt{\frac{4qV}{3m}}$$

$$= \sqrt{\frac{4(2 \times 1.60 \times 10^{-19})(750)}{3 \times 4u}}$$

$$= 2.195 \times 10^5$$

$$\approx 2.20 \times 10^5 \text{ m s}^{-1}$$

Change in speed

$$= v_2 - v_1$$

$$= 2.20 \times 10^5 - 1.58 \times 10^5$$

$$= 6.20 \times 10^3 \text{ m s}^{-1}$$

7(a)

The work function energy of zinc is lower than the energy of the ultraviolet photons,